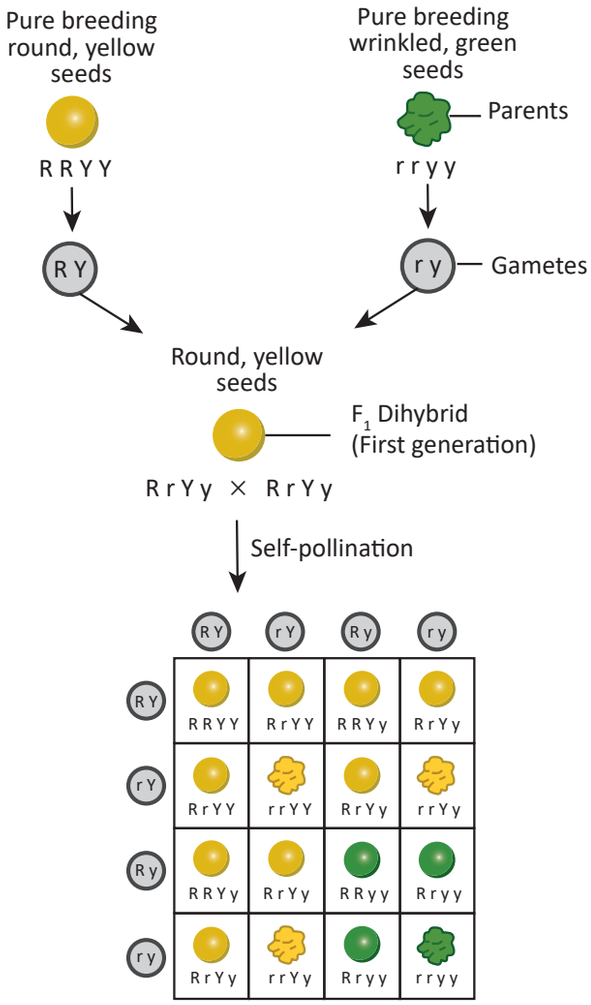


Phenotypic ratio of F_2 generation - 9:3:3:1

Round, Yellow - 9 Wrinkled, Yellow - 3
Round, Green - 3 Wrinkled, Green - 1

Figure 18.2 Dihybrid Cross

round yellow, **round-green**, **wrinkled yellow** and wrinkled-green seeds.

The ratio of each phenotype (or appearance) of seeds in the **F_2 generation is 9:3:3:1**. This is known as the **Dihybrid ratio**.

From the above results it can be concluded that the factors for each character or trait remain independent and maintain their identity in the gametes. The factors are independent to each other and pass to the offsprings (through gametes).

Results of a Dihybrid Cross:

Mendel got the following results from his dihybrid cross

1. **Four Types of Plants:** A dihybrid cross produced four types of F_2 offsprings in the ratio of 9 with two dominant traits, 3 with one dominant trait and one recessive trait, 3 with another dominant trait and another recessive trait and 1 with two recessive traits.
2. **New Combination:** Two new combinations of traits with round green and wrinkled yellow had appeared in the dihybrid cross (F_2 generation).

18.4 Mendel's Laws

Based on his experiments of monohybrid and dihybrid cross, Mendel proposed three important laws which are now called as Mendel's **Laws of Heredity**.

- **Law of Dominance:**

  "When two homozygous individuals with one or more sets of contrasting characters are crossed, the characters that appear in the F_1 hybrid are dominant and those that do not appear in F_1 are recessive characters".

- **Law of Segregation or Law of purity of gametes:**

  "When a pair of contrasting factors are brought together in a F_1 hybrid. The two factors of the allelic pair remain together without mixing and when gametes are formed, the two separate out, so that only one enters each gamete."

- **Law of independent assortment:**

  "In case of inheritance of two or more pairs of characters simultaneously, the

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